

Spring 2025

GRADE 6
MATHEMATICS

PTM0613_7.5

1. What is $|-7.5|$?

Enter your answer in the space.

PTM0605_P_4

2. Pavlova is a popular dessert in Australia. A manager at a restaurant wants to keep at least 25 servings of pavlova available for customers to order. Which inequality represents the number of servings of pavlova, p , available at the restaurant?

A. $p < 25$

B. $p \leq 25$

C. $p > 25$

D. $p \geq 25$

PTM0621_4,5,2

3. Which of the following questions are examples of statistical questions?

Select **all** the correct answers.

A. How tall is the Empire State Building?

B. What is your age?

C. What is the capital of Arkansas?

D. How many times have you been to the beach?

E. How much money do you earn each week?

M18_A288_44

4. An expression is shown.

$$3x^2 + 10y + 5z + 7$$

Evaluate the expression for $x = 3$, $y = 0$, and $z = 2$.

Enter your answer in the box.

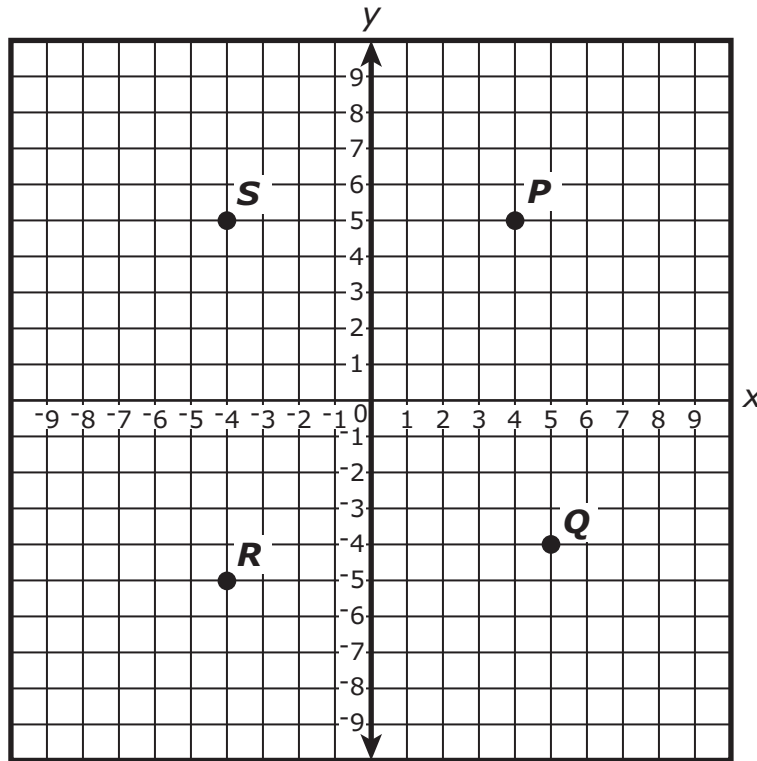
PTM0626_4

5. When plotting the point $(-4, 2)$ on the coordinate plane, the x -coordinate indicates that the point is how many units to the left of the origin?

Enter your answer in the box.

M18_B035_4

6. Four points are shown on the coordinate grid.



Which point is located at $(-4, 5)$?

- A. Point *P*
- B. Point *Q*
- C. Point *R*
- D. Point *S*

PTM0608_2

7. A rectangular plaque honoring war veterans is $\frac{3}{4}$ meter long and has an area of $\frac{1}{2}$ square meter. What is the width of the plaque?
- A. $\frac{3}{8}$ m
- B. $\frac{2}{3}$ m
- C. $\frac{3}{2}$ m
- D. $\frac{5}{4}$ m

PTM0602_4

8. Which expression is equivalent to $5(x - 7)$?
- A. $5x - 7$
- B. $x - 35$
- C. $5x + 35$
- D. $5x - 35$

PTM0611_22

9. What is the quotient?

$$11.88 \div 0.54$$

Enter your answer in the space.

PTM0612_5

- 10.** What is the greatest common factor of 10 and 75?

Enter your answer in the space.

PTM0601_214

- 11.** Evaluate the expression $7(x - 2)^3 + x^2$ when $x = 5$.

Enter your answer in the space.

PTM0610_215

- 12.** What is the quotient?

$$4,945 \div 23$$

Enter your answer in the space.

PTM0614_P_1

- 13.** Which point is a distance of 5 units away from (1, 3) and in Quadrant II?

A. $(-4, 3)$

B. $(1, -2)$

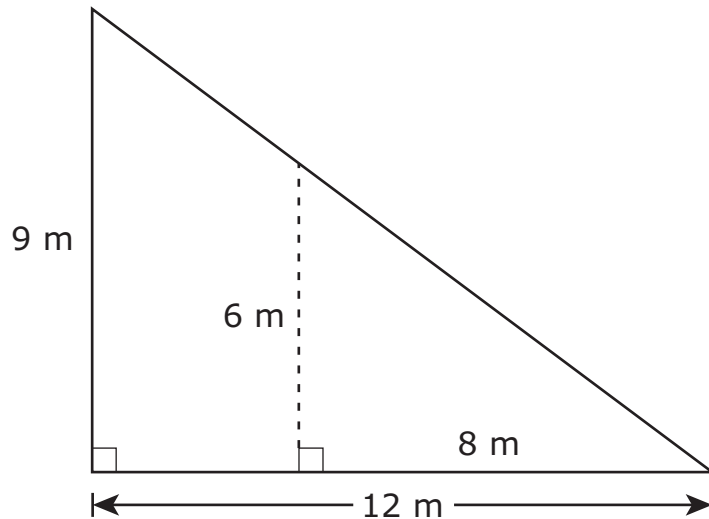
C. $(1, 8)$

D. $(6, 3)$

PTM0624_24:30

14. This question has two parts.

A vegetable garden is in the shape of a right triangle with legs measuring 12 meters and 9 meters. A farmer wants to divide the garden into two sections by installing a fence, as shown in the diagram.



The farmer will plant carrots in the triangular section of the garden and squash in the other section.

Part A

What is the area, in square meters, of the section of the garden where carrots will be planted?

Enter your answer in the space.

Part B

What is the area, in square meters, of the section of the garden where squash will be planted?

Enter your answer in the space.

PTM0604_3,2,1

- 15.** Which values make the inequality $2y + 3 > 3y$ true?

Select **all** the correct answers.

A. $y = 0$

B. $y = 1$

C. $y = 2$

D. $y = 3$

E. $y = 4$

PTM0618

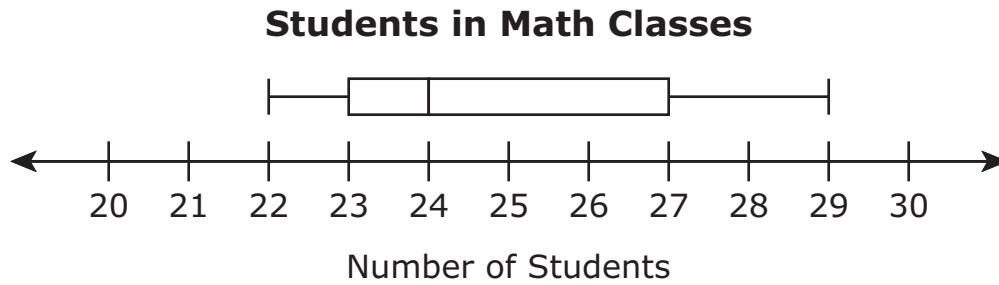
- 16.** Maya is hiking along a trail. She has hiked 4 miles in 2 hours. She has 5 more miles to hike to complete the trail. Maya continues to hike at the same rate.

- Explain how to determine Maya's hiking rate.
- How many total hours will it take Maya to complete the trail?
- Explain how you can use reasoned estimates to check your work.

Enter your answer in the space provided. Show all your work to support your answer.

6M23_074_2

- 17.** A principal recorded the number of students in each math class in the school. The box plot shown summarizes the data.

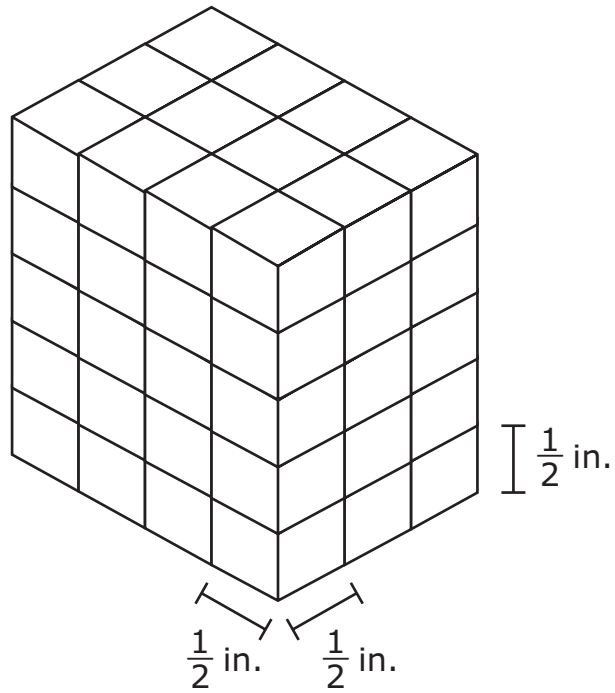


Based on the box plot, what is the median number of students in a math class in the principal's school?

- A.** 23
- B.** 24
- C.** 25
- D.** 27

PTM0607_7.5

- 18.** How many cubic inches is the volume of the rectangular prism shown?
Express your answer as a decimal.

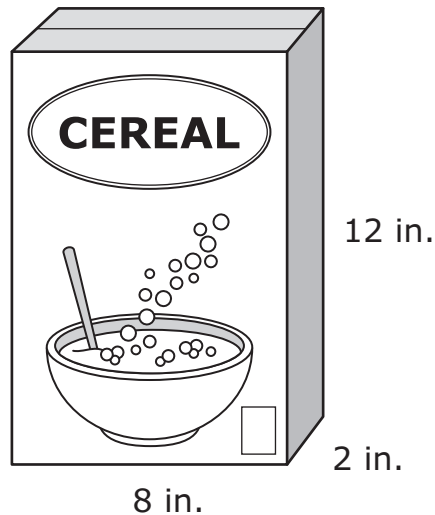


Enter your answer in the space.

PTM0622_:

19. This question has two parts.

A box of cereal with the dimensions shown is being shipped. It costs \$5.76 to ship the box of cereal. The box of cereal contains 6 cups of cereal.



Part A

A customer wants to know the cost for shipping each cubic inch of the box of cereal.

- Write an expression to find the volume of the box of cereal in cubic inches.
- What is the volume of the box of cereal?
- What is the cost for shipping each cubic inch of the box of cereal? Show your work.

Enter your answers in the space provided. Show all your work to support your answers.

Part B

A customer wants to know how many $\frac{1}{4}$ -cup servings of cereal there are in the box of cereal.

- Write an expression to find the number of $\frac{1}{4}$ -cup servings of cereal that are in the box of cereal.
- Use a model to find the number of $\frac{1}{4}$ -cup servings of cereal that are in the box of cereal. Show your work.
- How can you use the relationship between multiplication and division to find the number of $\frac{1}{4}$ -cup servings of cereal that are in the box of cereal?

Enter your answers in the space provided. Show all your work to support your answers.

PTM0603

20. Consider the expressions $-2(x - 5)$ and $10 - 2x$.

- What is the value of the expression $-2(x - 5)$ when $x = 3$?
- What is the value of the expression $10 - 2x$ when $x = 3$?
- Are the expressions $-2(x - 5)$ and $10 - 2x$ equivalent? How do you know?

Enter your answers in the space provided. Show all your work to support your answers.

PTM0627

- 21.** Samantha can bike at a constant rate of 12 miles per hour. Samantha's friend, Jake, can bike at a constant rate of 10 miles per hour.

Both Samantha and Jake start biking from the same point and at the same time. How much farther will Samantha have biked than Jake after 4 hours? Show your work or explain how you determined your answer.

Enter your answer in the space provided. Show all your work to support your answer.

PTM0609

- 22.** Jamie has $\frac{1}{2}$ cup of raisins. The diagram shows the amount of raisins Jamie has.



Jamie wants to know how many $\frac{1}{8}$ -cup servings of raisins she has.

- Write an expression to find the number of $\frac{1}{8}$ -cup servings of raisins that are in $\frac{1}{2}$ cup of raisins.
- How can the diagram representing $\frac{1}{2}$ cup of raisins be modified to show how many $\frac{1}{8}$ -cup servings of raisins are in $\frac{1}{2}$ cup of raisins?
- How many $\frac{1}{8}$ -cup servings of raisins are in $\frac{1}{2}$ cup of raisins? Explain using the relationship between multiplication and division.

Enter your answers in the space provided. Show all your work to support your answer.

23. Which table shows the correct conversions from pints and quarts to cups?

A.

Original Measurement	Number of Cups
2 pints	4
2 quarts	4

B.

Original Measurement	Number of Cups
2 pints	4
2 quarts	8

C.

Original Measurement	Number of Cups
2 pints	8
2 quarts	4

D.

Original Measurement	Number of Cups
2 pints	8
2 quarts	8

M18_A578_2

- 24.** An electronics store advertises 25% off the regular price of any stereo. What is the amount of the discount on a \$300 stereo?
- A.** \$25
 - B.** \$75
 - C.** \$225
 - D.** \$275

PTM0619_8

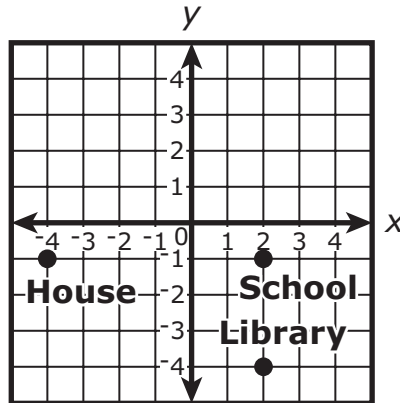
- 25.** Amari was making a banner for a Juneteenth parade. She placed a hook 25% of the way across the banner from left to right. The hook was 2 feet from the left side of the banner. What is the length, in feet, of Amari's banner?

Enter your answer in the box.

PTM0615_:

26. This question has two parts.

The map shows some locations in Paige's town. Each unit represents 1 mile.



Part A

Paige wants to know how far her house is from the school.

- Explain how to determine the distance from Paige's house to the school using absolute value.
- What is the distance from Paige's house to the school?

Enter your answer in the space provided. Show all your work to support your answer.

Part B

Paige wants to know how far the school is from the library.

- Explain how to determine the distance from the school to the library using absolute value.
- What is the distance from the school to the library?

Enter your answer in the space provided. Show all your work to support your answer.



Illinois MATH Assessment

Practice Item Answer Key

Grade 6 - Paper, Screen Reader, and Non-Screen Reader

The following pages include the answer key for all machine-scored items, followed by a sample response for the hand-scored item.

- The rubrics show sample student responses. Student responses other than that shown in the rubric may earn full or partial credit.
- Which responses to hand-scored items receive full or partial credit will be confirmed during range-finding (reviewing sets of real student work)
- If students make a computation error, they can still earn points for reasoning or modeling.

Item Number	Answer Key
1.	Student response is 7.5.
2.	D
3.	D, E, B
4.	Student response is 44.
5.	Student response is 4.
6.	D
7.	B
8.	D
9.	Student response is 22.
10.	Student response is 5.
11.	Student response is 214.
12.	Student response is 215.
13.	A
14.	Part A: 24 Part B: 30
15.	C, B, A



16.	See Rubric
17.	B
18.	Student response is 7.5.
19.	See Rubric
20.	See Rubric
21.	See Rubric
22.	See Rubric
23.	B
24.	B
25.	8
26.	See Rubric



#16 Rubric

Score	Description
3	Student response includes the following elements. • Reasoning component 1 = 1 point: Valid explanation for finding Maya's hiking rate. • Computation component = 1 point: Correct value for the total number of hours Maya hikes to complete the trail • Reasoning/Modeling component = 1 point: Valid explanation for how to use reasoned estimates to check the work. Sample Student Response: Maya has hiked 4 miles in 2 hours, which is a rate of 4 miles $\div$ 2 hours, or 2 miles per hour. It will take Maya 4.5 hours total to complete the trail. Maya has hiked 4 miles in 2 hours. She has 5 more miles to hike. It will take Maya slightly longer to hike 5 more miles than it took her to hike 4 miles. A reasonable estimate is that it will take Maya 3 hours to hike 5 more miles. So, a reasonable estimate is that it will take Maya 2 hours + 3 hours, or 5 total hours, to complete the trail. The estimate of 5 hours is close to the answer of 4.5 hours.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.



#19 Rubric

3 Point Constructed Response Rubric – Part A

Score	Description
3	Student response includes the following elements. • Modeling component = 1 point: Correct expression to find the volume of the box of cereal • Computation component = 1 point: Correct volume of the box of cereal • Computation component = 1 point: Correct cost for shipping each cubic inch of the box of cereal with work shown Sample Student Response: The box of cereal is in the shape of a rectangular prism. To find the volume of a rectangular prism, multiply the length by the width by the height. So, an expression to find the volume of the box of cereal is $8 \times 2 \times 12$. The volume of the box of cereal is $8 \times 2 \times 12 = 192$ cubic inches. Divide the cost to ship the box of cereal by the volume of the box of cereal. The cost for shipping each cubic inch of the box of cereal is $\\$5.76 \div 192 = \\0.03 per cubic inch. Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

3 Point Constructed Response Rubric – Part B

Score	Description
3	Student response includes the following elements. • Modeling component = 1 point: Correct expression to find the number of 1/4-cup servings of cereal there are in the box of cereal • Modeling component = 1 point: Correct description of how to use a model to find the number of 1/4-cup servings of cereal there are in the box of cereal



	Computation component = 1 point: Correct description of how to use the relationship between multiplication and division to find the number of $\frac{1}{4}$-cup servings of cereal there are in the box of cereal Sample Student Response: To find the number of $\frac{1}{4}$-cup servings of cereal there are in the box of cereal, divide the number of cups of cereal by the amount of one serving: $6 \div \frac{1}{4}$. Draw a model with 6 equal parts to represent the 6 cups of cereal. Divide each part into 4 equal parts to represent a serving size of $\frac{1}{4}$ cup. The model shows 24 equal parts, so there are twenty-four $\frac{1}{4}$-cup servings of cereal in the box of cereal. $6 \div \frac{1}{4} = 24$ because $24 \times \frac{1}{4} = 6$, so there are twenty-four $\frac{1}{4}$-cup servings of cereal in the box of cereal. Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

#20 Rubric

Score	Description
3	Student response includes the following elements. Computation component = 1 point: Correct value of the expression $-2(x - 5)$ when $x = 3$ Computation component = 1 point: Correct value of the expression $10 - 2x$ when $x = 3$ Modeling component = 1 point: Shows why the expressions $-2(x - 5)$ and $10 - 2x$ are equivalent Sample Student Response: When $x = 3$, $-2(x - 5) = -2(3 - 5) = -2(-2) = 4$. When $x = 3$, $10 - 2x = 10 - 2(3) = 10 - 6 = 4$. The expressions $-2(x - 5)$ and $10 - 2x$ are equivalent because when $x = 3$, or when $x = \text{any value}$, the expressions have the same value. (In other words, $-2(x - 5)$ and $10 - 2x$ are equivalent because $-2(x - 5) = -2x + 10 = 10 - 2x$.) Or other valid approaches are acceptable.



2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

#21 Rubric

Score	Description
3	Student response includes the following elements. • Modeling component = 1 point: Valid explanation or work shown for determining how many miles Samantha and Jake each rode their bike. • Modeling component = 1 point: Valid explanation or work shown for determining how much farther Samantha biked. • Computation component = 1 point: Correct value for the number of miles, 8 Sample Student Response: $12 \times 4 = 48$ $10 \times 4 = 40$ $48 - 40 = 8$ Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

#22 Rubric



Score	Description
3	Student response includes the following elements. • Modeling component = 1 point: Correct expression to find the number of $\frac{1}{8}$-cup servings of raisins that are in $\frac{1}{2}$ cup of raisins • Modeling component = 1 point: Correct description of how the model can be modified to show how many $\frac{1}{8}$-cup servings of raisins are in $\frac{1}{2}$ cup of raisins • Computation component = 1 point: Finds the correct number of $\frac{1}{8}$-cup servings of raisins in $\frac{1}{2}$ cup of raisins using the relationship between multiplication and division Sample Student Response: Divide the total amount of raisins by the amount of raisins in one serving. So, to find the number of $\frac{1}{8}$-cup servings of raisins that are in $\frac{1}{2}$ cup of raisins use the expression $(\frac{1}{2}) \div (\frac{1}{8})$. The diagram shows the $\frac{1}{2}$ cup of raisins that Jamie has. Divide it into 8 equal parts so that each part represents a $\frac{1}{8}$-cup serving. The diagram would show 4 of 8 parts shaded, which means there are four $\frac{1}{8}$-cup servings of raisins in $\frac{1}{2}$ cup of raisins. There are four $\frac{1}{8}$-cup servings of raisins in $\frac{1}{2}$ cup of raisins because $(\frac{1}{2}) \div (\frac{1}{8}) = 4$ since $\frac{1}{8} \times 4$ is $\frac{1}{2}$. Or other valid approaches are acceptable.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.



2 Point Constructed Response Rubric – Part A

Score	Description
2	Student response includes the following elements. • Reasoning component = 1 point: Explains how to determine the distance from Paige's house to the school using absolute value • Computation component = 1 point: Correct distance from Paige's house to the school Sample Student Response: Paige's house is located at $(-4, -1)$ and the school is located at $(2, -1)$. Since the y-coordinates are the same and the x-coordinates are opposites, find the sum of the absolute value of the x-coordinates. So, the distance from Paige's house to the school is $-4 + 2 = 4 + 2 = 6$ miles. Or other valid approaches are acceptable.
1	Student response includes 1 of the 2 elements.
0	Student response is incorrect or irrelevant.

2 Point Constructed Response Rubric – Part B

Score	Description
2	Student response includes the following elements. • Reasoning component = 1 point: Explains how to determine the distance from the school to the library using absolute value • Computation component = 1 point: Correct distance from the school to the library Sample Student Response: The school is located at $(2, -1)$ and the library is located at $(2, -4)$. Since the x-coordinates are the same and the y-coordinates are not opposites, find the difference of the absolute value of the y-coordinates. So, the distance from the school to the library is $-4 - -1 = 4 - 1 = 3$ miles. Or other valid approaches are acceptable.
1	Student response includes 1 of the 2 elements.
0	Student response is incorrect or irrelevant.